

Macroeconomics (ECON205B)

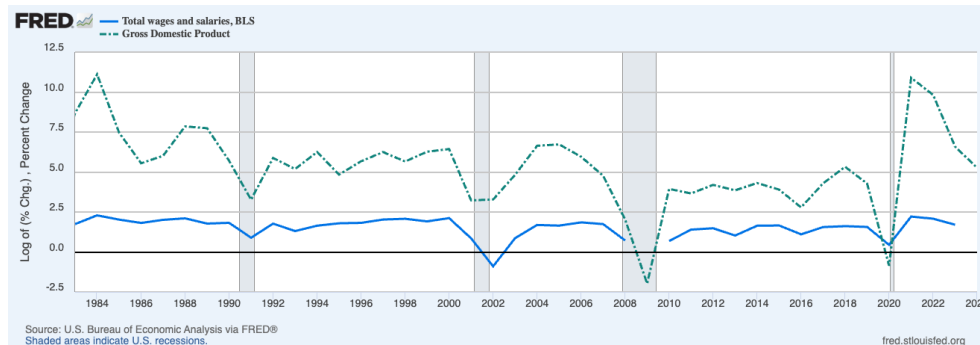
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Contents

1	Dynamic Programming Foundations	2
2	RBC: Growth, Detrending, Balanced Growth Path	2
3	RBC Quantitative Evaluation	3
4	Fiscal Policy Shocks	4
5	News Shock Model	6
6	Home Production in RBC	8
7	Money in Utility (MIU) Model	11
8	Cash in Advance (CIA) Model	13
9	New Keynesian Model	14

1 Dynamic Programming Foundations

Cake eating

Cake of size W_1 ; choose $\{c_t\}$ to consume. Lifetime utility $U = \sum_{t=1}^T \beta^{t-1} u(c_t)$, $\beta \in [0, 1]$, u concave, increasing, $\lim_{c \rightarrow 0} u'(c) = \infty$ Law of motion $W_{t+1} = W_t - c_t$

Sequential problem

$$\max_{\{W_{t+1}, c_t\}_{t=1}^T} \sum_{t=1}^T \beta^{t-1} u(c_t) \quad \text{st } W_1 = W_{T+1} + \sum_{t=1}^T c_t, \quad c_t, W_{T+1} \geq 0$$

Euler equation $u'(c_t) = \beta u'(c_{t+1})$ (the Inada condition rules out $c_t = 0$; $W_{T+1} \geq 0$ binds at 0, since leftover cake is wasted)

Value function

$$V_T(W_1) = \max_{\{W_{t+1}, c_t\}} \sum_{t=1}^T \beta^{t-1} u(c_t) \quad \text{s.t. same constraints} \quad (1)$$

Envelope: a marginal increase in cake raises lifetime utility by the marginal utility of any period, $V'_T(W_1) = \beta^{t-1} u'(c_t)$

Finite-horizon DP

Add a period 0 and reuse $V_T(W_1)$:

$$\max_{c_0} u(c_0) + \beta V_T(W_1) \quad \text{s.t. } W_1 = W_0 - c_0 \quad (2)$$

Principle of optimality: whatever is chosen at $t = 0$, continuation value is fully summarized by $V_T(W_1)$, so the problem separates into $t = 0$ and $t \geq 1$

Infinite-horizon DP (Bellman)

$$V(W) = \max_{c \in [0, W]} \{u(c) + \beta V(W - c)\} = \max_{W' \in [0, W]} \{u(W - W') + \beta V(W')\}, \quad (3)$$

W = state, W' = control. Stationarity: t drops out; only W matters. Solution is the fixed point V of the Bellman equation.

Existence/uniqueness For $V(s) = \max_{s' \in \Gamma(s)} \{u(s, s') + \beta V(s')\}$ with u real-valued, continuous, bounded, $\beta \in (0, 1)$, and $\Gamma(s)$ non-empty, compact, continuous, a unique V exists.

Bellman operator

$$T(X)(s) = \max_{s' \in \Gamma(s)} \{u(s, s') + \beta X(s')\} \quad (4)$$

T takes a guess X , maximizes, returns the update; any V with $T(V) = V$ solves the Bellman equation

Blackwell sufficiency $\Rightarrow$ contraction (i) Monotonicity: $X \geq Q \Rightarrow T(X) \geq T(Q)$ (ii) Discounting: $T(X + k) \leq T(X) + \beta k$, $\beta \in [0, 1)$. Together these make T a contraction, which has a unique fixed point reachable by iterating T from any guess (value function iteration).

2 RBC: Growth, Detrending, Balanced Growth Path

Growth cycles are not explained in a neoclassical model. This is because business cycle phenomena cannot be explained with a stochastic model, and therefore explains capital accumulation, but lacks variation in savings or employments. From RBC research, we can deduce a number of stylized facts, such that real output grows at a constant rate, real capital grows more or equal to the rate of labor, and real output and capital stock growth rates are similar. RBC attempts to better explain the dynamics we see in long run business cycles in the US, particularly the relationship of employment hours from modeled results and empirical data.

Production with labor-augmenting growth

$$Y_t = z_t K_{t-1}^\alpha (X_t H_t)^{1-\alpha}, \quad X_t = \gamma X_{t-1}, \quad \gamma > 1 \quad (5)$$

Preferences (KPR)

$$u(C_t, H_t) = \ln C_t - \varphi \frac{H_t^{1+\eta}}{1+\eta}, \quad (6)$$

η = inverse Frisch elasticity, φ = normalization pinning down SS hours. If $\sigma = 1$, $u(C, L) = \ln C + v(L)$. KPR with labor-augmenting growth is consistent with balanced growth because (i) IES is invariant to the scale of consumption and (ii) income and substitution effects cancel along the BGP.

Euler (log utility) and factor prices

$$\frac{1}{C_t} = \beta_t \left[\frac{1}{C_{t+1}} (r_{t+1} + 1 - \delta) \right], \quad r_t = \alpha \frac{Y_t}{K_{t-1}}, \quad w_t = (1 - \alpha) \frac{Y_t}{H_t} \quad (7)$$

With gross return $R_{t+1} \equiv r_{t+1} + 1 - \delta$, along the BGP $C_{t+1}/C_t = \beta R_{t+1}$

Detrended variables ($\tilde{V}_t \equiv V_t/X_t$, so $V_t = \tilde{V}_t X_t$; note $\tilde{K}_{t-1} = K_{t-1}/X_t$) Detrended equilibrium conditions (decentralized SME):

$$\begin{aligned} \frac{1}{\tilde{C}_t} &= \frac{\beta}{\gamma_t} \left[\frac{1}{\tilde{C}_{t+1}} (r_{t+1} + 1 - \delta) \right] && \text{(Euler)} \\ \frac{\tilde{W}_t}{\tilde{C}_t} &= \varphi H_t^\eta && \text{(labor supply)} \\ \tilde{Y}_t &= z_t \tilde{K}_{t-1}^\alpha H_t^{1-\alpha} && \text{(production)} \\ \gamma \tilde{K}_t &= (1 - \delta) \tilde{K}_{t-1} + \tilde{I}_t && \text{(capital)} \\ \tilde{C}_t + \tilde{I}_t &= \tilde{Y}_t && \text{(resource)} \end{aligned}$$

Growth shows up only through the β/γ discount in the Euler equation and the γ on capital accumulation

3 RBC Quantitative Evaluation

Steady state ($\bar{z} = 1$)

From the Euler equation, $1 = \beta(\bar{r} + 1 - \delta) \Rightarrow \bar{r} = \frac{1}{\beta} + \delta - 1$. Using $\bar{r} = \alpha \bar{Y}/\bar{K} = \alpha(\bar{K}/\bar{H})^{\alpha-1}$ and defining $\Omega \equiv \bar{K}/\bar{H}$:

$$\alpha \Omega^{\alpha-1} = \frac{1}{\beta} + \delta - 1 \Rightarrow \Omega = \left[\frac{\frac{1}{\beta} + \delta - 1}{\alpha} \right]^{\frac{1}{\alpha-1}} \quad (8)$$

Wage and labor supply give $\bar{W} = (1 - \alpha)\Omega^\alpha$ and $\bar{C} = (1 - \alpha)\Omega^\alpha/(\varphi \bar{H}^\eta)$ Capital accumulation gives $\bar{I} = \delta \bar{K}$; the resource constraint $\bar{C} + \delta \bar{K} = \bar{Y}$ then solves for hours:

$$\bar{H} = \left[\frac{1 - \alpha}{\varphi(1 - \delta \Omega^{1-\alpha})} \right]^{\frac{1}{1+\eta}} \quad (9)$$

Log-linearization

Log deviation $\hat{X}_t \equiv \ln(X_t/\bar{X})$ Tricks: (1) $X_t \approx \bar{X}(1 + \hat{X}_t)$; (2) $X_t Y_t \approx \bar{X}\bar{Y}(1 + \hat{X}_t + \hat{Y}_t)$; (3) $X_t^b \approx \bar{X}^b(1 + b\hat{X}_t)$ Pros: intuitive, fast. Cons: misses nonlinearities (large-shock responses are just scaled small-shock responses), can't handle non-convexities/kinks/discrete behavior

Log-linearized equilibrium conditions

$$\begin{aligned}
\widehat{C}_t &= {}_t[\widehat{C}_{t+1} - \beta \bar{r} \widehat{r}_{t+1}] && \text{(Euler)} \\
\eta \widehat{H}_t &= \widehat{W}_t - \widehat{C}_t && \text{(labor supply)} \\
\widehat{Y}_t &= \widehat{z}_t + \alpha \widehat{K}_{t-1} + (1 - \alpha) \widehat{H}_t && \text{(production)} \\
\widehat{W}_t &= \widehat{Y}_t - \widehat{H}_t, \quad \widehat{r}_t = \widehat{Y}_t - \widehat{K}_{t-1} \\
\widehat{K}_t &= (1 - \delta) \widehat{K}_{t-1} + \delta \widehat{I}_t && \text{(capital)} \\
\frac{\bar{C}}{\bar{Y}} \widehat{C}_t + \frac{\bar{I}}{\bar{Y}} \widehat{I}_t &= \widehat{Y}_t && \text{(resource)} \\
\widehat{z}_t &= \rho \widehat{z}_{t-1} + \varepsilon_t
\end{aligned}$$

Linear rational-expectations system

State $x_t = \widehat{K}_t$; jumps $y_t = [\widehat{C}_t, \widehat{I}_t, \widehat{Y}_t, \widehat{W}_t, \widehat{r}_t]'$; exogenous $z_t = \widehat{z}_t$

Business-cycle facts (HP filter)

Decompose $y_t = \tau_t + c_t$ (trend + cycle). The HP filter solves

$$\min_{\{\tau_t\}} \sum_{t=1}^T (y_t - \tau_t)^2 + \lambda \sum_{t=2}^{T-1} [(\tau_{t+1} - \tau_t) - (\tau_t - \tau_{t-1})]^2, \quad (10)$$

first term keeps the trend near data, second penalizes trend curvature

Frisch elasticity

With $u = \ln C_t - \varphi H_t^{1+\nu}/(1+\nu)$: $\lambda_t = 1/C_t$, $u_2 = -\varphi H_t^\nu$ Intratemporal FOC $-\varphi H_t^\nu + \lambda_t W_t = 0$ gives $H_t = (\lambda_t W_t / \varphi)^{1/\nu}$, so holding λ_t fixed,

$$\frac{\partial \ln H_t}{\partial \ln W_t} = \frac{1}{\nu} \quad \text{(Frisch elasticity)} \quad (11)$$

Technology $\ln z_t = \rho \ln z_{t-1} + \varepsilon_t$ is recovered as a Solow residual, $\ln z_t = \ln Y_t - \alpha \ln K_{t-1} - (1 - \alpha) \ln H_t$

4 Fiscal Policy Shocks

- In RBC, technology shocks only shift the labor demand curve. We want a model that shifts the labor supply curve.
- In the model, govt finances expenditures with capital & labor:
 - if $\text{rev} > \text{exp} \Rightarrow$ households get a transfer;
 - if $\text{rev} < \text{exp} \Rightarrow$ households pay a lump-sum tax
- Fiscal shocks help us understand the low association between hours worked & productivity

Transmission

- Capital & labor income tax change the effective rental & wage rates
- An increase in govt spending lowers the transfer via the govt budget constraint
- An increase in govt spending shifts the labor supply curve, leading to lower wages and a higher number of hours worked (negative income effect)
- An increase in govt spending increases productivity, mainly via more hours worked
- Govt spending is not 1-to-1 with a change in output because govt crowds out investment and consumption (gains from govt are offset by a decrease in investment, consumption & savings)

How to solve

- Only decentralized approach (sequential or recursive)
- Production side of economy is the same as in the standard RBC model
- Differences from standard RBC: (1) aggregate state variables include Y_t, τ_t, g_t ; (2) govt budget constraint is satisfied; (3) the resource constraint includes g_t

Model set-up (general)

(i) **Sequential** Households max utility:

$$\left[\sum_{t=0}^{\infty} \beta^t u(C_t + \pi g_t, h_t) \right] \quad (12)$$

Household budget:

$$C_t + i_t \leq r_t k_{t-1} + w_t h_t + \xi_t - \tau_t(r_t - \delta)k_{t-1} - \ell_t w_t h_t \quad (13)$$

where π = weight on pref for govt spending ($0 \leq \pi \leq 1$)

τ_t = capital income tax rate

ℓ_t = labor income tax rate

ξ_t = govt transfer ($\xi > 0 \Rightarrow$ transfer, else lump-sum tax)

govt budget constraint: $\tau_t(r_t - \delta)k_{t+1} + \ell_t w_t h_t = g_t + \xi_t$ % VERIFY k_{t+1} vs k_{t-1}

per capita transfers: $\xi_t = \tau_t(r_t - \delta)k_{t-1} + \ell_t w_t h_t - g_t$

govt resource constraint: $C_t + i_t + g_t = Y_t$

Exogenous states: $v_t = [z_t, g_t, \tau_t, \ell_t]'$, where $v_{t+1} = (1 - \rho_v)\bar{v} + \rho_v v_t + \varepsilon_{t+1}$ (AR(1) shock process)

Four potential shocks

TFP shock: $\ln(z_t) = \rho_z \ln(z_{t-1}) + \zeta_{z,t}, \quad \zeta_{z,t} \sim N(0, \tau_z^2)$

govt spending: $\ln(g_t) = (1 - \rho_g) \ln(\bar{g}) + \rho_g \ln g_{t-1} + \zeta_{g,t}$

capital income tax: $\ln(\tau_t) = (1 - \rho_\tau) \ln(\bar{\tau}) + \rho_\tau \ln \tau_{t-1} + \zeta_{\tau,t}$

labor income tax: $\ln(\ell_t) = (1 - \rho_\ell) \ln(\bar{\ell}) + \rho_\ell \ln \ell_{t-1} + \zeta_{\ell,t}$

Household problem

$$\max_{C_t, i_t, h_t} \sum \beta^t u(C_t, h_t) \quad \text{s.t. household budget \& law of capital motion}$$

Firms:

$$\max_{k_t^d, h_t^d} [z_t F(k_t^d, h_t^d) - r_t k_t^d - w_t h_t^d], \quad \text{state vars } v = \{k_{t-1}, z_t, \tau_t, \ell_t, g_t\}$$

Household policy $f(x)$: $C(v, k_{t-1}), h(v, k_{t-1}), i(v, k_{t-1})$ Firm policy $f(x)$: $k^d(v), h^d(v)$

Sequential market equilibrium

label=(1),nosep prices: $w(v), r(v)$

lbbel=(2),nosep LCM: $k_t = m(k_{t-1}, z_t, \ell_t, g_t)$

lcbel=(3),nosep govt transfer: $\xi_t(k_{t-1}, z_t, \tau_t, \ell_t, g_t) = \xi_t(v_t)$

ldbel=(4),nosep govt budget constraint holds

lebel=(5),nosep all markets clear: $h_t = H_t = H_t^d, \quad k_t = K_t = K_t^d,$

$$C_t + i_t + g_t = C_t + i_t + g_t = z_t H_t^\alpha K_t^\alpha \quad \% \text{ VERIFY exponents}$$

To solve (sequential, decentralized): Lagrange Collapse the law of capital motion into the household budget constraint; use summation over s (in addition to existing t):

$$= \sum_{s=0}^{\infty} \beta^{t+s} \left[u(C_{t+s} + g_{t+s}(\pi), h_{t+s}) - \lambda_{t+s} [C_{t+s} + k_{t+s} - (1-\delta)k_{t+s-1} - r_t k_{t+s-1} - w_{t+s} h_{t+s} + \tau_{t+s}(r_{t+s} - \delta)k_{t+s-1} + \ell_{t+s} w_{t+s} h_{t+s} - \xi_{t+s}] \right] \quad (14)$$

FOCs:

$$\begin{aligned}\frac{\partial}{\partial C_t} &= u_c - \lambda_t = 0 \Rightarrow \lambda_t = u_c(C_{t+s} + g_{t+s}\pi, h_{t+s}) \\ \frac{\partial}{\partial h_t} &= u_h + \lambda_t(w_t - \ell_t w_t) \Rightarrow -u_h = \lambda_t(w_t - \ell_t w_t) \\ \frac{\partial}{\partial k_t} &= -\lambda_t - \lambda_{t+1}[(1 - \delta) - r_{t+1} + \tau_{t+1}(r_{t+1} - \delta)]\beta\end{aligned}$$

Labor-supply FOC (combine $\partial/\partial C_t$ and $\partial/\partial h_t$):

$$-u_h = u_c w_t(1 - \ell_t) \quad (15)$$

Intertemporal (Euler) equation (sub $\partial/\partial C_t$ into $\partial/\partial k_t$):

$$u_{c,t} = u_{c,t+1}[(1 - \delta) - r_{t+1} + \tau_{t+1}r_{t+1} - \delta\tau_{t+1}]\beta \quad (16)$$

so

$$u_{c,t} = \beta u_{c,t+1}[1 - \delta - r_{t+1} + \tau_{t+1}(r_{t+1} - \delta)] \quad \% \text{ VERIFY sign pattern} \quad (17)$$

Other conditions: $w_t = F_h(k_{t-1}, H_t)$, $k_t = (1 - \delta)k_{t-1} + i_t$
 $r_t = F_k(k_{t-1}, H_t)$, $Y_t = z_t F(k_{t-1}, H_t)$
 $C_t + i_t + g_t = Y_t$

To solve (recursive) Set up value function; define state variables $S = \{K, z, \tau, \ell, g\}$:

$$\begin{aligned}V(K, S) &= \max_{C, i, k} \left\{ u(C + \pi g, h) + \beta V(K', S') \right\} && \text{(household value fn)} \\ &= \max_{z} zK^d H^d - rK^d - wH^d && \text{(firm's problem)}\end{aligned}$$

Household policy: $C(K, S)$, $i(K, S)$, $h(K, S)$ Firm policy: $K^d(S)$, $H^d(S)$

Prices $r(S)$, $w(S)$; law of capital motion $K' = m(S)$; govt transfer $\xi(S)$ Markets clear, govt budget constraint satisfied & firms/houses optimize $\Rightarrow$ recursive competitive equilibrium:

$$h = H = H^d, \quad k = K = K^d, \quad C + i + g = C + i + g = z F(K, H)$$

5 News Shock Model

- Purpose: use the RBC with augmentation that allows for news about the future to create a boom today, as RBC is constrained.
- Introduces: (1) variable levels of capital K utilization; (2) adjustment costs for investment choices; (3) weak, short-run wealth effect on labor supply.
- Tobin's Q : η_t/λ_t is the value of an additional unit of installed capital, in terms of C_t .
- We look for a response by aggregate variables in response to news of a shock.

Transmission

- An expected future increase in A or z implies investment will rise in the future; with adjustment costs it is optimal to smooth investment, so investment also increases today
- Capital becomes cheaper because of higher investment today, so utilization increases
- MPL provides an incentive to supply more labor (MPL increases)
- Output only increases in aggregate if the MPL increase is greater than the rise in disutility of labor (wealth effect)
- $\gamma = 0 \Rightarrow$ GHH preferences: wealth effect is null & workers want to work more
- $\gamma = 1 \Rightarrow$ KPR preferences: real future wages rise & workers want to work less today

Model set-up

Household utility:

$$U = \sum_{t=0}^{\infty} \beta^t \left[\frac{(C_t - \Psi N_t^\theta \chi_t)^{1-\sigma} - 1}{1-\sigma} \right] \quad (18)$$

where $\chi_t = C_t^\gamma \chi_{t-1}^{(1-\gamma)}$, and C_t = consumption

N_t = hours worked

γ = strength of wealth effect

Three cases for γ : (1) $\gamma = 0 \Rightarrow$ GHH pref, negative comovement between C_t, N_t ; $\gamma = 1 \Rightarrow$ KPR pref, N_t^θ not consistent;
 $0 \leq \gamma \leq 1 \Rightarrow$ balanced growth path, C_t, χ_t grow at constant rate

With 3 cases for γ you get different FOCs & interpretations of “boom”; consider it when interpreting results.

$Y_t = A_t(u_t K_{t-1})^{1-\kappa} N_t^\kappa$	production function (output), with shock	% VERIFY exponents
$C_t + \frac{i_t}{z_t} = Y_t$	resource constraint, with shock	
$K_t = \left[1 - \psi\left(\frac{i_t}{i_{t-1}}\right)\right] i_t + [1 - \delta(u_t)] K_{t-1}$	law of motion of capital	
$\psi\left(\frac{i_t}{i_{t-1}}\right) = \frac{K}{2} \left(\frac{i_t}{i_{t-1}} - 1\right)^2$	investment adjustment cost	
$\delta(u_t) = \delta_0 + \delta_1(u_t - 1) + \frac{\delta_2}{2}(u_t - 1)^2$	functional form of depreciation rate	

where A_t = productivity shock, u_t = capital utilization rate, z_t = investment-specific tech shock, K = parameter describing propensity of firms to smooth investment.

To solve — (1) Social planner allocation

Set up Lagrange with 3 constraints: first a constraint on χ_t , then a combined constraint for Y_t (production, resource), finally a third combined constraint of adjustment cost and law of motion of capital. We do not sub capital into the resource constraint because of investment adjustment costs. (1) $C_t^\gamma \chi_{t-1}^{1-\gamma} - \chi_t$

(2) $A_t(u_t K_{t-1})^{1-\kappa} N_t^\kappa - \left[C_t + \frac{i_t}{z_t}\right]$

(3) $\left[1 - \psi\left(\frac{i_t}{i_{t-1}}\right)\right] i_t + [1 - \delta(u_t)] K_{t-1}$

$$= \sum_{t=0}^{\infty} \beta^t \left[\frac{(C_t - \Psi N_t^\theta \chi_t)^{1-\sigma} - 1}{1-\sigma} + \mu_t [C_t^\gamma \chi_{t-1}^{1-\gamma} - \chi_t] + \lambda_t [A_t(u_t K_{t-1})^{1-\kappa} N_t^\kappa - C_t + \frac{i_t}{z_t}] + \eta_t \left[\left(1 - \psi\left(\frac{i_t}{i_{t-1}}\right)\right) i_t + (1 - \delta(u_t)) K_{t-1} \right] \right] \quad (19)$$

FOCs:

$$\begin{aligned} \frac{\partial}{\partial C_t} = 0 &= (C_t - \Psi N_t^\theta \chi_t)^{-\sigma} + \mu_t \gamma C_t^{\gamma-1} \chi_{t-1}^{1-\gamma} - \lambda_t \\ \frac{\partial}{\partial N_t} = 0 &= -(C_t - \Psi N_t^\theta \chi_t)^{-\sigma} \theta N_t^{\theta-1} + \lambda_t (1 - \kappa) A_t(u_t K_{t-1})^{-\kappa} N_t^{\kappa-1} \quad \% \text{ VERIFY} \\ \frac{\partial}{\partial \chi_t} = 0 &= -(C_t - \Psi N_t^\theta \chi_t)^{-\sigma} - \mu_t + \beta [\mu_{t+1} (1 - \gamma) C_{t+1}^\gamma \chi_t^{-\gamma}] \\ \frac{\partial}{\partial u_t} = 0 &= \lambda_t (1 - \kappa) A_t(u_t K_{t-1})^{-\kappa} N_t^\kappa - \eta_t (\delta'(u_t) K_{t-1}) \\ \frac{\partial}{\partial K_t} = 0 &= \beta^{t+1} (\lambda_{t+1} (A_t(u_t K_t))^{1-\kappa} N_{t+1}^\kappa (1 - \kappa) + \eta_{t+1} (1 - \delta(u_{t+1}))) - \beta^t \eta_t \\ \frac{\partial}{\partial i_t} = 0 &= \lambda_t \left(-\frac{1}{z_t}\right) + \eta_t \left\{ \left(1 - \psi\left(\frac{i_t}{i_{t-1}}\right)\right) + \psi'\left(\frac{i_t}{i_{t-1}}\right) \left(\frac{i_t}{i_{t-1}}\right) \right\} + \beta^2 \left[\eta_{t+1} \left(\psi'\left(\frac{i_{t+1}}{i_t}\right) \left(\frac{i_{t+1}}{i_t}\right)^2\right) \right] \end{aligned}$$

Rearranged:

$$\begin{aligned} \text{from } \frac{\partial}{\partial C_t} : \quad \lambda_t &= (C_t - \Psi N_t^\theta \chi_t)^{-\sigma} + \mu_t (\gamma C_t^{\gamma-1} \chi_{t-1}^{1-\gamma}) \\ \text{from } \frac{\partial}{\partial \chi_t} : \quad \mu_t &+ (C_t - \Psi N_t^\theta \chi_t)^{-\sigma} = \beta [\mu_{t+1} (1 - \gamma) C_{t+1}^\gamma \chi_t^{-\gamma}] \\ \text{from } \frac{\partial}{\partial N_t} : \quad (C_t - \Psi N_t^\theta \chi_t)^{-\sigma} \theta N_t^{\theta-1} &= \lambda_t (Y_t / N_t^\kappa), \quad Y_t / N_t^\kappa = A_t(u_t K_{t-1})^{1-\kappa} \\ \text{from } \frac{\partial}{\partial u_t} : \quad \lambda_t (1 - \kappa) Y_t / u_t &= \eta_t \delta'(u_t) K_{t-1}, \quad A_t(u_t K_{t-1})^{1-\kappa} N_t^\kappa = Y_t / u_t \end{aligned}$$

Interpretation:

- From $\partial/\partial u_t$: $\eta_t \delta'(u_t) K_{t-1} = \text{LHS} \Rightarrow$ shadow price of capital multiplied by the marginal cost of increasing the capital utilization rate.
- From $\partial/\partial u_t$: $\lambda_t(1 - \kappa)Y_t/u_t = \text{RHS} \Rightarrow$ marginal return on an increase in K utilization.
- From $\partial/\partial K_t$ (reduce by eliminating discount rate):

$$\eta_t = \beta_t [\lambda_{t+1}(1 - \kappa)Y_{t+1}/K_t + \eta_{t+1}(1 - \delta(u_{t+1}))] \quad (20)$$

So the marginal return on using capital stock (LHS) equals the marginal return on increasing capital stock, less its depreciation value, in discounted expectations.

- From $\partial/\partial i_t$:

$$\frac{\lambda_t}{z_t} = \eta_t \left\{ 1 - \frac{K}{2} \left(\frac{i_t}{i_{t-1}} - 1 \right)^2 + K \left(\frac{i_t}{i_{t-1}} - 1 \right) \right\} + \beta_t \left[\eta_{t+1} \left(\frac{i_{t+1}}{i_t} \right) \left(\frac{i_{t+1}}{i_t} \right)^2 \right] \quad \% \text{ VERIFY} \quad (21)$$

To solve — (2) Decentralized approach (allocation is the same)

Households:

$$\begin{aligned} & \max_{C_t, \chi_t, i_t, N_t, u_t} \sum \beta^t [(C_t - \Psi N_t^\theta \chi_t)^{1-\sigma} / (1 - \sigma)] \\ \text{s.t. } & \chi_t = C_t^\gamma \chi_{t-1}^{1-\gamma} \\ & K_t = (1 - \delta(u_t))K_{t-1} + (1 - \psi(\frac{i_t}{i_{t-1}}))i_t \\ & C_t + i_t = w_t N_t + r_t K_{t-1} u_t \end{aligned}$$

Households choose $C_t, \chi_t, i_t, N_t, u_t$ (they choose utilization rate u_t when renting K)

Firms:

$$\max_{K_t^d, N_t^d} [A_t(u_t K_t^d)^{1-\kappa} (N_t^d)^\kappa - r_t u_t K_t^d - w_t N_t^d] \quad (22)$$

Set up Lagrange, take FOCs (like social planner method). Firm FOCs:

$$\begin{aligned} \frac{\partial}{\partial K_t^d} &= 0 = (1 - \kappa) A_t (u_t K_t^d)^{-\kappa} N_t^{d\kappa} - r_t u_t \\ \frac{\partial}{\partial N_t^d} &= 0 = \kappa A_t (u_t K_t^d)^{1-\kappa} N_t^{d\kappa-1} - w_t \end{aligned}$$

Impose market-clearing: $K_t^d = K_{t-1} = K_t, N_t^d = N_t = N_t$

- rewrite: $(1 - \kappa)(Y_t/K_t^d) = r_t u_t$
- multiply both sides by $K_{t-1} (= K_t^d)$: $(1 - \kappa)Y_t = r_t u_t K_{t-1}$
- use household FOC from $\partial/\partial u_t$ where $\lambda_t r_t = \eta_t \delta'(u_t)$ to replace r_t with $\eta_t \delta'(u_t)$
- now: $\lambda_t(1 - \kappa)(Y_t/u_t) = \eta_t \delta'(u_t) K_{t-1} \Rightarrow$ equivalent to social planner (provides constraint on capital utilization)
- reduce to: $\lambda_t r_t = \eta_t \delta'(u_t)$

6 Home Production in RBC

- Incorporates home production into the standard RBC model, so agents may substitute between producing goods at home (for consumption) & in the market (for wages); home production uses time and capital to make non-tradeable goods
- Agents substitute across sectors (home, market) and over time
- Allows the labor-leisure tradeoff to have a wealth effect, showing how individuals change their tradeoff as a result of a change in lifetime income

Transmission

- Business & home investments are negatively correlated in model, despite empirical data showing they move together, likely because capital can be allocated across sectors
- Key result: hours worked is more volatile than standard RBC, as agents can sub in & out of market activities, using home production as a buffer

Model set-up

Utility function for households:

$$u(C, \ell) = \log(C) + A \log(\ell) \quad (23)$$

with

$$C_t = [a C_{m,t}^e + (1-a) C_{H,t}^e]^{1/e}$$

$$\ell_t = 1 - h_{m,t} - h_{H,t}$$

Constant elasticity of substitution: $1/(1-e)$, where $C_{H,t}$ = home consumption, $C_{m,t}$ = market consumption

$$f(z_{m,t}, K_{m,t-1}, h_{m,t}) = e^{z_{m,t}} K_{m,t-1}^\theta h_{m,t}^{1-\theta} \quad \text{market productivity}$$

$$g(z_{H,t}, K_{H,t-1}, h_{H,t}) = e^{z_{H,t}} K_{H,t-1}^\eta h_{H,t}^{1-\eta} \quad \text{home productivity}$$

Shocks to home or market:

$$z_{m,t+1} = \rho z_{m,t} + \varepsilon_{m,t}$$

$$z_{H,t+1} = \rho z_{H,t} + \varepsilon_{H,t}$$

$\text{corr}(\varepsilon_{m,t}, \varepsilon_{H,t}) \neq 0$ (not always)

$$K_{m,t} = (1-\delta)K_{m,t-1} + i_{m,t} \quad \text{LCM for market}$$

$$K_{H,t} = (1-\delta)K_{H,t-1} + i_{H,t} \quad \text{LCM for home}$$

$$C_{m,t} + i_{m,t} + i_{H,t} = f(z_{m,t}, K_{m,t-1}, h_{m,t}) \quad \text{resource constraints}$$

$$C_{H,t} = g(z_{H,t}, K_{H,t-1}, h_{H,t})$$

To solve

Note there are four ways to solve: social planner & decentralized, sequential and recursive. Solve social planner both ways & decentralized as sequential.

Two-stage problem

- 1 Maximize consumption C_t given the utility form & a CES equation for C_t , where $C_t = (C_H + C_M)$
- 2 Find equilibrium via either method using constrained optimization

(1)

Part (1): consumption aggregation

$$\max_{C_1, C_2} C = [a C_1^e + (1-a) C_2^e]^{1/e} \quad \text{s.t. } E = p_1 C_1 + p_2 C_2$$

$$= (a C_1^e + (1-a) C_2^e)^{1/e} - \lambda (E - p_1 C_1 - p_2 C_2)$$

FOCs:

$$\frac{\partial}{\partial C_1} = 0 = \frac{1}{e} (a C_1^{e-1} + (1-a) C_2^{e-1}) a e C_1^{e-1} - \lambda p_1 \Rightarrow a (C/C_1)^{1-e} = \lambda p_1$$

$$\frac{\partial}{\partial C_2} = 0 = \frac{1}{e} (a C_1^{e-1} + (1-a) C_2^{e-1}) (1-a) C_2^{e-1} - \lambda p_2 \Rightarrow (1-a) (C/C_2)^{1-e} = \lambda p_2$$

Combining both:

$$\frac{\lambda p_1}{\lambda p_2} = \frac{a (C/C_1)^{1-e}}{(1-a) (C/C_2)^{1-e}} = \frac{p_1}{p_2} \Rightarrow \left(\frac{p_1}{p_2} \right)^{-1/(1-e)} \left(\frac{a}{1-a} \right)^{-1/(1-e)} = \frac{C_1}{C_2} \quad (24)$$

The ratio C_1/C_2 is the expenditure ratio used in the tradeoff between home & market consumption; look at value of e for substitutability:

label=(1),nosep $e \rightarrow 0$: Cobb–Douglas

lbbel=(2),nosep $e \rightarrow 1$: perfect substitutes

lcbel=(3),nosep $e \rightarrow \infty$: Leontief, perfect complements

Part (2): social planner, sequential Substitute $K_{m,t}, K_{H,t}$ into the resource constraint, for two constraints; use entire form (of utility & production) so you can see where variables become relevant

$$= \sum \beta^t \left[u(C_t, \ell_t) - \lambda_t (C_{m,t} + K_{m,t} - (1-\delta)K_{m,t-1} + K_{H,t} - (1-\delta)K_{H,t-1} - e^{z_{m,t}} K_{m,t-1}^\theta h_{m,t}^{1-\theta}) - \mu_t (C_{H,t} - e^{z_{H,t}} K_{H,t-1}^\eta h_{H,t}^{1-\eta}) \right] \quad (25)$$

with $u(C_t, \ell_t) = \ln(C_t) + A \ln(1 - h_{m,t} - h_{H,t})$

FOCs:

$$\begin{aligned} \frac{\partial}{\partial C_{m,t}} = 0 &= \frac{1}{C_t} \cdot \frac{1}{e} \cdot C_t^{(1/e-1)} \dots - \lambda_t \Rightarrow \lambda_t = \frac{1}{C_t} \cdot \frac{1}{e} (a C_{m,t}^e + (1-a) C_{H,t}^e)^{(1/e-1)} (a e C_{m,t}^{e-1}) \\ \frac{\partial}{\partial C_{H,t}} = 0 &= C_t^{-e} (1-a) C_{H,t}^{e-1} - \mu_t \Rightarrow \mu_t = C_t^e (1-a) C_{H,t}^{e-1} \\ \frac{\partial}{\partial h_{m,t}} = 0 &= -A(\ell_t)^{-1} + \mu_t (\theta \exp(z_{m,t}) K_{m,t-1}^\theta \cdot (1-\eta) h_{H,t}^{-\eta}) = 0 \\ \frac{\partial}{\partial h_{H,t}} = 0 &= -A(1/\ell_t) + \lambda_t (1-\theta) (\exp(z_{m,t})) K_{m,t-1}^\theta h_{m,t}^{-\theta} = 0 \\ \frac{\partial}{\partial K_{m,t}} = 0 &= -\lambda_t \beta^t + \beta^{t+1} [\lambda_{t+1} \exp(z_{m,t+1}) \theta K_{m,t}^{\theta-1} h_{m,t+1}^{1-\theta} + 1 - \delta] = 0 \\ \frac{\partial}{\partial K_{H,t}} = 0 &= -\lambda_t \beta^t + \beta^{t+1} [\mu_{t+1} \exp(z_{H,t+1}) \eta K_{H,t}^{\eta-1} + \lambda_{t+1} (1-\delta)] = 0 \end{aligned}$$

From these FOCs, rearrange & interpret:

- $\lambda_t = C_t^{-e} a C_{m,t}^{e-1} \Rightarrow$ marginal utility of market consumption
- $\mu_t = C_t^{-e} (1-a) C_{H,t}^{e-1} \Rightarrow$ marginal utility of home consumption
- rearranging $\partial/\partial h_{H,t}$: marginal disutility of home labor equals, in equilibrium, the marginal return to home labor:
 $A/\ell_t = \mu_t (1-\eta) g_{H,t}/h_{H,t}$
- marginal utility of market labor equals the marginal return to market labor: $A/\ell_t = \lambda_t (1-\theta) f_{m,t}/h_{m,t}$
- $\lambda_t = \beta_t [\lambda_{t+1} f_{m,t+1}/K_{m,t}(\theta) + 1 - \delta] \Rightarrow$ marginal return on market investment = marginal utility of market consumption
- $\lambda_t = \beta_t [\delta \lambda_{t+1} - \lambda_{t+1} + \mu_{t+1}(\eta) g_{H,t+1}/K_{H,t+1}] \Rightarrow$ LHS is MU of market consumption, RHS is marginal product of home capital $\times$ marginal utility of consuming at home

Part (2): social planner, recursive Social planner is uniquely constrained by technology for investment. Set up value function:

$$V(I_{-1}, K, z) = u(C, H) + \beta[V(I, K', z')] \quad (26)$$

Problem set-up:

$$\max_{I, K', C, H} [u(C, H) + \beta[V(I, K', z')]] \quad \text{s.t. } C + I = zF(K, H), \quad K' = (1-\delta)K + \frac{1}{2}I + \frac{1}{2}I_{-1} \quad \% \text{ VERIFY} \quad (27)$$

Note this problem also has the additional feature that investment is allocated in two sectors of investment. State variables: $\{K, I_{-1}\} = S$, where state variables refer to whether or not the variable is predetermined & less about the time period.

Part (2): decentralized, sequential Begin with household optimization & firms' optimization. Recall production functions, law of capital motion(s), and resource constraints. Use consumption agg def for C_t & household budget:

$$C_{m,t} + i_{m,t} + i_{H,t} \leq w_t h_{m,t} + r_t K_{m,t-1} \quad (28)$$

Firms' problem: $(K_{m,t})^\theta (H_{m,t})^{1-\theta} - \rho \Rightarrow$ substitutability of labor & capital in market

$$\max_{K_{m,t}^d, H_{m,t}^d} [e^{z_{m,t}} \cdot \rho - r_t K_{m,t}^d - w_t h_{m,t}^d] \quad (29)$$

Substitute LCM into household budget constraint, eliminate it from equation. Using 2 constraints, form the Lagrange:

$$= \max \sum \beta^t \left[\ln C_t + A \ln(1 - h_{m,t} - h_{H,t}) - \lambda_t (C_{m,t} + K_{m,t} - (1-\delta)K_{m,t-1} + K_{H,t} - (1-\delta)K_{H,t-1} - w_t h_{m,t} - r_t K_{m,t-1}) - \mu_t (C_{H,t} - e^{z_{H,t}} K_{H,t-1}^\eta h_{H,t}^{1-\eta}) \right] \quad (30)$$

FOCs:

$$\begin{aligned} \frac{\partial}{\partial C_{m,t}} = 0 &= a e C_{m,t}^{e-1} C_t^{1-e} (C_t)^{-1} = \lambda_t \\ \frac{\partial}{\partial C_{H,t}} = 0 &= (1-a) e C_{H,t}^{e-1} C_t^{1-e} (C_t)^{-1} = \mu_t \\ \frac{\partial}{\partial K_{m,t}} = 0 &= \beta^t (-\lambda_t + \beta [\lambda_{t+1} (1-\delta) + \beta \lambda_{t+1} r_{t+1}]) \Rightarrow \lambda_t = \beta [\lambda_{t+1} (1-\delta) + r_{t+1}] \\ \frac{\partial}{\partial K_{H,t}} = 0 &= \beta^t (-\lambda_t + \beta \lambda_{t+1} (1-\delta) + \beta \mu_t + e^{z_{H,t+1}} (\eta) K_{H,t}^{\eta-1} h_{H,t+1}^{1-\eta}) = 0 \end{aligned}$$

rearranged: $\lambda_t = {}_t\beta\lambda_{t+1}(1-\delta) + {}_t\beta\mu_t + e^{z_{H,t+1}}(\eta)K_{H,t}^{\eta-1}h_{H,t+1}^{1-\eta}$

$$\begin{aligned}\frac{\partial}{\partial h_{m,t}} = 0 &= \beta^t(-A/(1-h_{m,t}-h_{H,t})-\lambda_t w_t) = 0 \Rightarrow A(1-h_{m,t}-h_{H,t})^{-1} = \lambda_t w_t \\ \frac{\partial}{\partial h_{H,t}} = 0 &= \beta^t\left(-A/(1-h_{m,t}-h_{H,t})-\mu_t e^{z_{H,t}}K_{H,t-1}^\eta(1-\eta)h_{H,t}^{-\eta}\right) = 0 \\ \text{rearrange: } A(1-h_{m,t}-h_{H,t})^{-1} &= \mu_t \cdot (1-\eta)g(z_{H,t}, h_{H,t}, K_{H,t})\end{aligned}$$

where λ_t = marginal utility of market consumption, μ_t = marginal value of home production (governing consumption optimization)

$$\frac{1}{C_t} = {}_t\beta(C_{t+1})^{-1}[(1-\delta) + r_{t+1}] \Rightarrow \text{Euler equation (sub value of } \lambda_t \text{ from } \partial/\partial C_{m,t} \text{ into } \partial/\partial K_{m,t}) \quad (31)$$

7 Money in Utility (MIU) Model

- Real money balances directly enter the utility function; agents hold some positive quantity of money in equilibrium
- Partial equilibrium model where we hold constraints on households, then maximize utility with money (m_t) in the utility function
- Choice variables: consumption, labor, capital, real bond & real money holdings
- Key relationship: between real money holdings and the interest rate
- **Friedman rule:** the optimal amount of money is achieved when the nominal interest rate is zero, because there is no opportunity cost of holding money
 - if $i_t > 0$: households keep a finite amount of money, demand is also finite
 - if $i_t < 0$: optimal rate of inflation is deflationary, equivalent to the real return on capital
 - if $i_t = 0$: eliminate wedge (inefficiency) between private & social cost of money holding
- **Fisher equation:** $1 + i_t = (1 + r_t)(1 + \pi_t)$; relationship between nominal interest rate, real interest rate & rate of inflation, approximated as $i_t \approx r_t + \pi_t$
- To get this equation: get FOC for capital accumulation from optimizing utility; reduce FOC to Euler equation; approximate using logs of both, and define growth in consumption as $g_c = \frac{C_{t+1} - C_t}{C_t}$

Transmission

- Money holdings yield direct utility; higher inflation reduces those holdings, so inflation generates welfare losses (although a deflationary rate is optimal in SS).
- To interpret the steady state & growth dynamics, assume consumption & labor are additively separable from money in the utility function. When this holds we have:
 - **Level effect (neutrality):** a one-off increase in the money supply only increases utility through a direct channel, and therefore only affects a change in price levels.
 - **Growth effect (superneutrality):** change in the growth rate of money does NOT affect real variables. If it does not hold, then it does increase the rate of inflation.
 - Value of money is the price of a unit of money (λ_t/p_t), and is equal to the expected PV utility of the return on money.

How to solve

Given a utility function which households want to maximize, households choose labor, leisure, capital, bonds (if present) & money. We will (1) deflate budget & resource constraint to real values; (2) set up a Lagrange for household with constraints.

Model set-up

Households maximize utility:

$$\sum_{i=0}^{\infty} \beta^i U(C_{t+i}, m_{t+i}, \ell_{t+i}) \quad (32)$$

where ℓ_t = leisure, C_t = consumption, m_t = real money holdings = M_t/P_t (M_t = nominal money balance, P_t = price level)

Budget constraint (nominal):

$$P_t C_t + P_t K_t + B_t + M_t = P_t w_t(1 - \ell_t) + (r_t + 1 - \delta)P_t K_{t-1} + (1 + i_{t-1})B_{t-1} + M_{t-1} + P_t \tau_t \quad (33)$$

where B_t = nominal bond holdings $\Rightarrow B_t/P_t = b_t$ real bond holdings, w_t = wage, τ_t = real lump-sum transfer to households. Households must make choices in $(t - 1)$ affecting optimal choices in t .

(a) Deflate nominal budget constraint Divide the BC by P_t on LHS, RHS:

$$\frac{P_t C_t}{P_t} + \frac{P_t K_t}{P_t} + \frac{B_t}{P_t} + \frac{M_t}{P_t} = \frac{P_t w_t(1 - \ell_t)}{P_t} + \frac{(r_t + 1 - \delta)P_t K_{t-1}}{P_t} + \frac{(1 + i_t)B_{t-1}}{P_t} + \frac{M_{t-1}}{P_t} + \frac{P_t \tau_t}{P_t} \quad (34)$$

$$= C_t + K_t + b_t + m_t = w_t(1 - \ell_t) + (r_t + 1 - \delta)K_{t-1} + (1 + i_t)\frac{b_{t-1}}{(1 + \pi_t)} + \frac{m_{t-1}}{(1 + \pi_t)} + \tau_t \quad (35)$$

where $P_t/P_{t-1} = (1 + \pi_t) \Rightarrow$ net inflation rate. So on RHS, b_{t-1}, m_{t-1} are in (real + inflation) terms.

(b) FOCs for sequential market equilibrium via Lagrange

$$= {}_t \sum_{i=0}^{\infty} \beta^i \left(U(C_{t+i}, m_{t+i}, \ell_{t+i}) + \lambda_{t+i} (C_t + K_t + b_t + m_t - w_t(1 - \ell_t) - (r_t + 1 - \delta)K_{t-1} + (1 + i_{t-1})b_{t-1}/(1 + \pi_t) + m_{t-1}/(1 + \pi_t) + \tau_t) \right) \quad \% \text{ (all)} \quad (36)$$

FOCs:

$$\begin{aligned} \frac{\partial}{\partial C_t} = 0 &= U_c(\cdot) - \lambda_{t+i} \Rightarrow U_c(\cdot) = \lambda_t \\ \frac{\partial}{\partial \ell_t} = 0 &= U_\ell(\cdot) - \lambda_{t+i} w_t \Rightarrow U_\ell(\cdot) = \lambda_t w_t \\ \frac{\partial}{\partial K_t} = 0 &= -\lambda_{t+i}(1) - \lambda_{t+i+1}(r_{t+1} + 1 - \delta)\beta \Rightarrow [\beta \lambda_{t+1}(r_{t+1} + 1 - \delta)] = \lambda_t \\ \frac{\partial}{\partial b_t} = 0 &= -\lambda_{t+i}(1) - \lambda_{t+i+1}(1/(1 + \pi_{t+1}))(1 + i_t) \Rightarrow \beta[\lambda_{t+1}(\frac{1+i_t}{1+\pi_{t+1}})] = \lambda_t \\ \frac{\partial}{\partial m_t} = 0 &= U_m(\cdot) - \lambda_{t+i}(\cdot) - \beta[\lambda_{t+i+1}(1/(1 + \pi_t))] \Rightarrow \beta[\lambda_{t+1}(1/(1 + \pi_{t+1}))] = \lambda_t + U_m(\cdot) \end{aligned}$$

Bond Euler equation (from $\partial/\partial b_t$), LHS = marginal utility of discounted consumption = RHS = discounted expected marginal return on savings from bond holding:

$$\lambda_{t+i} = \beta \left[\lambda_{t+i+1} \left(\frac{1+i_{t+i}}{1+\pi_{t+i+1}} \right) \right] \quad (37)$$

From $\partial/\partial m_t$, marginal utility of consumption less the marginal utility of real money balances (LHS) equals the return to discounted marg utility (RHS):

$$\lambda_{t+i} - U_m(\cdot) = \beta[\lambda_{t+i+1}/(1 + \pi_{t+i+1})] \quad (38)$$

rearrange:

$$\lambda_{t+i} = U_m(\cdot) + \beta[\lambda_{t+i+1}/(1 + \pi_{t+i+1})] \quad (39)$$

where RHS is the marginal return on holding real money balance

Fisher equation from $\partial U/\partial b_t, \partial U/\partial m_t$ & knowing:

label=(1),nosep $(1 + \pi_{t+1}) = P_{t+1}/P_t \Rightarrow$ inflation rate in period $t + 1$

lbbel=(2),nosep $(1 + r)(1 + \pi) = (1 + i)$

lcbel=(3),nosep i = nominal interest rate, r = real interest rate (rental rate of capital), π = inflation rate

ldbel=(4),nosep $1/P_t$ = value of money in time $= t$

Value of money in NPV Taking $\partial U/\partial m_t$, iterate forward. Use (1) and put in FOC:

$$\begin{aligned}\lambda_t &= U_m(\cdot) + \beta[\lambda_{t+1}/P_{t+1} \cdot P_t] \cdot (1/P_t) \quad [\text{sub} \times (1/P_t)] \\ \frac{\lambda_t}{P_t} &= U_m(\cdot) + \beta[\lambda_{t+1}/P_{t+1}] \\ &= U_m(\cdot) + \beta \left[\frac{\partial U/\partial m_{t+1}}{P_{t+1}} \right] + \beta^2 \left[\frac{\partial U/\partial m_{t+2}}{P_{t+2}} \right] + \dots \\ &= {}_t \left[\sum_{i=0}^{\infty} \beta^i (\partial U/\partial m_{t+i}/P_{t+i}) \right] \\ \frac{1}{P_t} &= (1/\lambda_t) {}_t \left[\sum_i \beta^i \left(\frac{\partial U/\partial m_{t+i}}{P_{t+i}} \right) \right]\end{aligned}$$

Plug $\partial/\partial C_t$ into $\partial/\partial m_t$ to find the marginal rate of substitution for money & consumption, where $\lambda_t = \partial U/\partial C_t$:

$$\frac{\partial U/\partial m_t}{\partial U/\partial C_t} = \frac{i_t}{1+i_t} \implies \frac{U_m(\cdot)}{U_c(\cdot)} = \frac{i_t}{1+i_t} \quad (40)$$

LHS: MRS of m_t, C_t ; RHS: opportunity cost of holding money

8 Cash in Advance (CIA) Model

- Similar to MIU, but without direct effect on utility; instead money is held to finance certain types of goods that require cash
- Time assumption is important: the goods market opens first, then the asset market
- With positive opportunity cost, only the amount of money for purchases is held
- Households do not have labor, so the model is abstracted from labor supply
- Timing matters: if there was uncertainty in future, households may carry too much or too little cash to make purchases in next period

Transmission

- Inflation acts as a tax on goods where cash is required for purchase, creating a wedge between the MRT of technology & MRS faced by consumers
- In optimality, inflation should equal zero (to eliminate cost of holding money) & derivation of the asset pricing equation for money is possible

How to solve

Household's decision problem (no future uncertainty, drop expectations):

$$V(\omega_t, m_{t-1}) = \max \{u(C_t) + \beta V(\omega_{t+1}, m_t)\} \quad \text{value function optimization} \quad (41)$$

subject to:

- (1) budget constraint (nominal): $P_t \omega_t = P_t(K_{t-1} + (1 - \delta))P_t K_{t-1} + M_{t-1} + \tau_t + (1 + i_{t-1})B_{t-1}$ % VERIFY
- (2) CIA constraint (nominal): $P_t C_t \leq M_{t-1} + \tau_t$
- (3) resource constraint (nominal): $P_t \omega_t \geq P_t C_t + P_t K_t + M_t + B_t$

where P_t = aggregate price level, ω_t = nominal resource endowment (asset class), $a_t = m_t + b_t \Rightarrow$ real asset holdings, money & bonds

(1) Deflate constraints to real values where $R_t = (1 + i_t)/(1 + \pi_{t+1}) \Rightarrow$ real interest, $i_t/\pi_{t+1} \Rightarrow$ opp cost of holding money, $1/P_t \Rightarrow$ price deflator for nominal values So (3) becomes:

real resource constraint: $\omega_t \geq C_t + m_t + b_t + K_t$

real budget constraint: $\omega_{t+1} = f(K_t) + (1 - \delta)K_t + \tau_{t+1} + R_t(a_t) - \frac{i_t}{\pi_{t+1}}m_t$ (iterate forward one period)

real CIA constraint: $C_t \leq m_{t-1}/P_t + \tau_t/P_t = m_{t-1}/\pi_t + \tau_t$

Lagrange

$$= \max \left[u(C_t) + \beta V(f(K_t) + (1-\delta)K_t + \tau_{t+1} + R_t(a_t) - \frac{i_t}{\pi_{t+1}}(m_t), m_t) + \lambda_t(C_t + m_t + b_t + K_t - \omega_t) + \mu_t(m_{t-1}/\pi_t + \tau_t - C_t) \right] \quad (42)$$

C_t, K_t, b_t, m_t are choice variables; m_{t-1} is a state variable (given)

FOCs:

$$\begin{aligned} \frac{\partial}{\partial C_t} = 0 &= u_c(\cdot) - \lambda_t - \mu_t = 0 \Rightarrow u_c(\cdot) = \lambda_t + \mu_t \\ \frac{\partial}{\partial K_t} = 0 &= \beta V_K(\omega_{t+1}, m_t)(f'(K_t) + (1-\delta)) - \lambda_t = 0 \Rightarrow \beta V_K(\omega_{t+1}, m_t)[f'(K_t) + 1 - \delta] = \lambda_t \\ \frac{\partial}{\partial b_t} = 0 &= \beta V_a(\omega_{t+1}, m_t)R_t - \lambda_t = 0 \Rightarrow \beta V_a(\omega_{t+1}, m_t)R_t = \lambda_t \\ \frac{\partial}{\partial m_t} = 0 &= \beta V_m(\omega_{t+1}, m_t)(-i_t/\pi_{t+1}) + \beta V_{m2}(\omega_{t+1}, m_t) - \lambda_t + 0 = 0 \\ &\Rightarrow \lambda_t = \beta[V_{m1}(\cdot) \times -i_t/\pi_{t+1} + V_{m2}(\cdot)] \end{aligned}$$

Envelope theorem

$$\begin{aligned} \text{(a)} \quad V_\omega(\omega_t, m_{t-1}) &= \lambda_t \Rightarrow V_\omega(\omega_{t+1}, m_t) = \lambda_{t+1} \\ \text{(b)} \quad V_m(\omega_t, m_{t-1}) &= \mu_t(\pi_t)^{-1} \Rightarrow V_m(\omega_{t+1}, m_t) = \frac{\mu_{t+1}}{\pi_{t+1}} \end{aligned}$$

So now $\beta V_K(\omega_{t+1}, m_t) = \beta V_a(\omega_{t+1}, m_t) = \lambda_{t+1}$ Take this result & substitute into $\partial/\partial K_t$:

$$\lambda_t = \lambda_{t+1}[f'(K_t) + 1 - \delta] \quad (43)$$

where λ_t = marg utility of wealth, μ_t = marg value of holding money in real terms, $\lambda_t + \mu_t$ = marg value of consumption (as seen from $\partial/\partial C_t$)

Asset pricing equation for money

$$\lambda_t = \beta[\lambda_{t+1} + \mu_{t+1}/\pi_{t+1}] \Rightarrow \frac{1}{P_t} = (1/\lambda_t)\beta \left[\frac{\lambda_{t+1} + \mu_{t+1}}{P_{t+1}} \right] \quad (44)$$

If we iterate λ_t/P_t forward, we get $\sum_{i=0}^{\infty} \beta^i \frac{u(C_{t+i})}{P_{t+i}} \Rightarrow$ price of money in terms of utility (PV)

Can also see $u_c(\cdot) = \lambda_t + \mu_t = \lambda(1 + \mu) = \lambda(1 + i_t) \geq \lambda \Rightarrow$ so positive nominal interest rates lower consumption, as seen in the relationship between the marginal utility of consumption & nominal interest rates

9 New Keynesian Model

- NK extends the monetary model for policy analysis: nominal rates set by a central bank, money endogenous. Firms are monopolistically competitive with sticky prices (Calvo). Households buy differentiated goods, so consumption is a CES aggregate over the continuum.
- Transmission: an IS curve linking the output gap to the real rate and expected future output; a Phillips curve (NKPC) in which real marginal cost drives forward-looking inflation; determinacy requires a Taylor rule with inflation feedback > 1 . Without stickiness, NK collapses to RBC.

Common block: demand and the price index

Consumption aggregator and expenditure minimization:

$$C_t = \left(\int_0^1 C_t(i)^{\frac{\epsilon-1}{\epsilon}} di \right)^{\frac{\epsilon}{\epsilon-1}}, \quad \min_{\{C_t(i)\}} \int_0^1 P_t(i) C_t(i) di \text{ s.t. } C_t \geq \bar{C}_t \quad (45)$$

FOC $\Rightarrow$ demand curve and price index:

$$C_t(i) = \left(\frac{P_t(i)}{P_t} \right)^{-\epsilon} C_t, \quad P_t = \left(\int_0^1 P_t(i)^{1-\epsilon} di \right)^{\frac{1}{1-\epsilon}}, \quad (46)$$

and $\int_0^1 P_t(i)C_t(i) di = P_t C_t$ Household FOCs (CRRA C , separable labor) give the Euler and labor-supply conditions

$$c_t = {}_t c_{t+1} - \frac{1}{\sigma}(i_t - {}_t \pi_{t+1} - \rho) + \frac{1}{\sigma}({}_t z_{t+1} - z_t), \quad w_t - p_t = \sigma c_t + \varphi n_t \quad (47)$$

Firm technology $Y_t(i) = A_t N_t(i)^{1-\alpha}$; real marginal cost in logs $\psi_t = w_t - a_t + \alpha n_t - \ln(1-\alpha)$ The two approaches below differ only in the Calvo bookkeeping and in whether the NKPC is written in marginal cost or in the output gap

Approach A — Galí convention (θ = fraction that cannot reset; NKPC in the output gap)

Each period a fraction $1 - \theta$ of firms reset; θ keep last period's price. Price index dynamics:

$$P_t^{1-\varepsilon} = \theta P_{t-1}^{1-\varepsilon} + (1-\theta)(P_t^*)^{1-\varepsilon} \Rightarrow \pi_t = (1-\theta)(p_t^* - p_{t-1}) \quad (48)$$

Optimal reset price A resetting firm maximizes discounted real profit with the demand curve as constraint:

$$\max_{P_t^*} \sum_{k=0}^{\infty} \theta_t^k \left\{ A_{t,t+k} \frac{1}{P_{t+k}} [P_t^* Y_{t+k|t} - C_{t+k}(Y_{t+k|t})] \right\}, \quad A_{t,t+k} = \beta^k \frac{U_{c,t+k}}{U_{c,t}} \quad (49)$$

FOC (bracketed term): $[P_t^* - \mathcal{M}\Psi_{t+k|t}]Y_{t+k|t} = 0$ with frictionless markup $\mathcal{M} = \varepsilon/(\varepsilon - 1)$ Full FOC:

$$\sum_{k=0}^{\infty} \theta_t^k \left\{ A_{t,t+k} \frac{1}{P_{t+k}} [P_t^* - \mathcal{M}\Psi_{t+k|t}] Y_{t+k|t} \right\} = 0 \quad (50)$$

Flexible-price limit ($\theta = 0$): $P_t^* = \mathcal{M}\Psi_t$; as $\varepsilon \rightarrow \infty$, $\mathcal{M} \rightarrow 1$ (perfect substitutes)

Log-linearized reset price Around the zero-inflation SS ($\Lambda = \beta^k$, $P_t^*/P_{t+k} = 1$, $P = \mathcal{M}\Psi$):

$$\widehat{p}_t^* = (1 - \beta\theta) \sum_{k=0}^{\infty} (\beta\theta)^k {}_t [\widehat{\Psi}_{t+k|t}] \Rightarrow p_t^* = \mu + (1 - \beta\theta) \sum_{k=0}^{\infty} (\beta\theta)^k {}_t [\psi_{t+k|t}] \quad (51)$$

NKPC Combining reset price and index dynamics:

$$\pi_t = \beta_t \pi_{t+1} - \lambda \widehat{\mu}_t, \quad \lambda = \frac{(1-\theta)(1-\beta\theta)}{\theta} \Theta \quad (\Theta = 1 \text{ if } \alpha = 0), \quad (52)$$

where $\widehat{\mu}_t = \ln(\mu_t/\mu)$ is the average markup gap. Higher θ (stickier) $\Rightarrow$ flatter curve ($\partial\lambda/\partial\theta < 0$). Iterating, $\pi_t = -\lambda \sum_{k=0}^{\infty} \beta_t^k \widehat{\mu}_{t+k}$.

Markup gap $\rightarrow$ output gap With $\psi_t = w_t - a_t + \alpha n_t - \ln(1-\alpha)$, $w_t - p_t = \sigma y_t + \varphi n_t$, and $n_t = (y_t - a_t)/(1-\alpha)$:

$$\mu_t = -\left(\sigma + \frac{\varphi+\alpha}{1-\alpha}\right) y_t + \frac{1+\varphi}{1-\alpha} a_t + \ln(1-\alpha), \quad \widehat{\mu}_t = -\left(\sigma + \frac{\varphi+\alpha}{1-\alpha}\right) \widetilde{y}_t, \quad (53)$$

$\widetilde{y}_t \equiv y_t - y_t^n$ Hence the NKPC in the output gap:

$$\boxed{\pi_t = \beta_t \pi_{t+1} + \kappa \widetilde{y}_t, \quad \kappa = \lambda \left(\sigma + \frac{\varphi+\alpha}{1-\alpha} \right)} \quad (54)$$

Dynamic IS From the Euler equation with $y_t = c_t$ and natural rate $r_t^n = \rho + (1 - \rho_z)z_t + \sigma({}_t y_{t+1}^n - y_t^n)$:

$$\boxed{\widetilde{y}_t = {}_t \widetilde{y}_{t+1} - \frac{1}{\sigma}({}_t i_t - {}_t \pi_{t+1} - r_t^n)} \quad (55)$$

Iterating, $\widetilde{y}_t = -\frac{1}{\sigma} \sum_{k=0}^{\infty} {}_t (r_{t+k} - r_{t+k}^n)$ Closed with a Taylor rule $i_t = \rho + \phi_\pi \pi_t + \phi_y \widetilde{y}_t + v_t$; $\phi_\pi > 1$ delivers determinacy

Approach B — handwritten convention (ω = fraction that can adjust; NKPC in marginal cost)

Here $\omega \in [0, 1]$ is the share of firms that can reset each period, and the discount weight in the profit sum is written $\omega^i \Delta_{i,t+i}$ with $\Delta_{i,t+i} = \beta^i (C_{t+i})^{-\sigma} / C_t$ Firm cost minimization gives real marginal cost $\psi_t = (w_t/P_t)/z_t$

Profit maximization With the demand curve substituted in:

$$\max_{P_t^*} \sum_{i=0}^{\infty} \omega^i \Delta_{i,t+i} \left[(P_t^*/P_{t+i})^{1-\varepsilon} - \psi_{t+i} (P_t^*/P_{t+i})^{-\varepsilon} \right] C_{t+i} \quad (56)$$

Reducing the FOC and substituting $\Delta_{i,t+i}$:

$$\frac{P_t^*}{P_t} = \left(\frac{\varepsilon}{\varepsilon-1} \right) \frac{\sum_t \omega^i \beta^i C_{t+i}^{1-\sigma} \psi_{t+i} (P_t/P_{t+i})^{-\varepsilon}}{\sum_t \omega^i \beta^i C_{t+i}^{1-\sigma} (P_t/P_{t+i})^{1-\varepsilon}} \quad (57)$$

Adjusting firms set P_t^* ; non-adjusters carry P_{t-1} Price index law of motion:

$$P_t^{1-\varepsilon} = (1-\omega)(P_t^*)^{1-\varepsilon} + \omega P_{t-1}^{1-\varepsilon} \quad (58)$$

Flexible-price special case $\omega = 0$ (all adjust)

$$P_t^*/P_t = \left(\frac{\varepsilon}{\varepsilon-1} \right) \psi_t = \mu \psi_t, \quad \mu > 1, \quad (59)$$

so $z_t/\mu = w_t/P_t$, and with labor supply $\chi N_t^\varphi/C_t^{-\sigma} = z_t/\mu$, $N_t = Y_t/z_t$, flexible output $y_t^f = (1/\mu\chi)^{1/(\varphi+\sigma)} z_t^{(1+\varphi)/(\varphi+\sigma)}$

NKPC in marginal cost Linearizing the reset-price and index equations:

$$\boxed{\pi_t = \beta_t \pi_{t+1} + \tilde{\kappa} \hat{\psi}_t, \quad \tilde{\kappa} = \frac{(1-\omega)(1-\beta\omega)}{\omega}} \quad (60)$$

$\hat{\psi}_t$ = deviation of real marginal cost Using labor supply and linearizing to convert to the gap gives $\hat{\psi}_t = (\sigma + \varphi)\hat{y}_t - \dots$, and at $\omega = 0$, $\hat{\psi}_t = (\sigma + \varphi)(\hat{y}_t - \hat{y}_t^f)$

Linearized IS

$$\hat{y}_t = \hat{y}_{t+1} - \frac{1}{\sigma}(\hat{i}_t - \pi_{t+1}), \quad \tilde{y}_t = \tilde{y}_{t+1} - \frac{1}{\sigma}(\hat{i}_t - \pi_{t+1}) - u_t \quad (61)$$

Reconciling the two

Same model, different labels The mapping is $\theta_{(A)} \leftrightarrow \omega_{(B)}$ playing opposite roles: A's θ is the non-adjusting share, B's ω is the adjusting share, so B's " ω " equals A's " $1 - \theta$." The NKPC slopes are the same object written on the two conventions:

$$\kappa_{(A)} = \underbrace{\frac{(1-\theta)(1-\beta\theta)}{\theta}}_{\lambda} \left(\sigma + \frac{\varphi+\alpha}{1-\alpha} \right), \quad \tilde{\kappa}_{(B)} = \frac{(1-\omega)(1-\beta\omega)}{\omega} \quad (62)$$

A writes the forcing variable as the output gap (having already substituted $\hat{\mu}_t = -(\sigma + \frac{\varphi+\alpha}{1-\alpha})\tilde{y}_t$); B leaves it as real marginal cost $\hat{\psi}_t$. Since $\hat{\psi}_t$ is proportional to $\tilde{y}_t$ up to the same slope factor, the two NKPCs coincide once you (i) match adjusting/non-adjusting shares and (ii) convert marginal cost to the gap. Divine coincidence ($\pi_t = 0, \tilde{y}_t = 0$ via $i_t = r_t^n$) holds in both absent a cost-push term u_t .